PHÂN TÍCH THỐNG KÊ NHIỀU CHIỀU (MAT3452)

Bài tập tuần 9: Hồi quy tuyến tính nhiều biến

Trần Đăng Quang Minh - MSV: 23000141

```r
# II. Lựa chọn mô hình phù hợp
# 1. Hai mô hình lồng nhau
mod <- lm(mpg ~ cyl, data = mtcars)
mod1 <- lm(mpg ~ cyl + disp, data = mtcars)
mod2 <- lm(mpg ~ cyl + disp + hp, data = mtcars)
# P-value > 0.05 nên là việc thêm biến disp không có ý nghĩa
anova(mod, mod1) # H1: Sự thêm biến disp có ý nghĩa
```

```
## Analysis of Variance Table
##
## Model 1: mpg ~ cyl
## Model 2: mpg ~ cyl + disp
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1     30 308.33
## 2     29 270.74  1    37.594 4.0268 0.05419 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# P-value > 0.05 nên là việc thêm biến hp không có ý nghĩa
anova(mod1, mod2) # H1: Sự thêm biến hp có ý nghĩa
```

```
## Analysis of Variance Table
##
## Model 1: mpg ~ cyl + disp
## Model 2: mpg ~ cyl + disp + hp
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     29 270.74
## 2     28 261.37  1    9.3709 1.0039  0.325
```

```r
# 2. So sánh chỉ số AIC và BIC của các mô hình, có thể thông qua hàm step
# Sử dụng trực tiếp hàm AIC() và BIC() để tính ra giá trị chỉ số trực tiếp
# AIC và BIC càng thấp là mô hình càng tốt
# -> theo đánh giá qua AIC và BIC thì mod1 tốt nhất
AIC(mod)
```

```
## [1] 169.3064
```

```r
AIC(mod1)
```

```
## [1] 167.1456
```

```r
AIC(mod2)
```

```
## [1] 168.0184
```

```r
BIC(mod)
```

```
## [1] 173.7036
```

```r
BIC(mod1)
```

```
## [1] 173.0086
```

```r
BIC(mod2)
```

```
## [1] 175.3471
```

```r
# Sử dụng hàm step để tìm ra mô hình phù hợp nhất
# theo tiêu chuẩn trong các mô hình 1 biến, 2 biến,..., m biến
# Hàm step(): Giá trị mặc định của k = 2, ứng với chỉ số AIC ban đầu
# Nếu thay k = log(n) ta sẽ thu được chỉ số BIC
lm(mpg ~ 1, data = mtcars)
```

```
##
## Call:
## lm(formula = mpg ~ 1, data = mtcars)
##
## Coefficients:
## (Intercept)
##       20.09
```

```r
lm(mpg ~ cyl, data = mtcars)
```

```
##
## Call:
## lm(formula = mpg ~ cyl, data = mtcars)
##
## Coefficients:
## (Intercept)          cyl
##      37.885       -2.876
```

```r
lm(mpg ~ cyl + disp, data = mtcars)
```

```
##
## Call:
## lm(formula = mpg ~ cyl + disp, data = mtcars)
##
## Coefficients:
## (Intercept)          cyl         disp
##    34.66099     -1.58728     -0.02058
```

```r
lm(mpg ~ ., data = mtcars) # Theo tất cả các biến
```

```
##
## Call:
## lm(formula = mpg ~ ., data = mtcars)
##
## Coefficients:
## (Intercept)          cyl         disp           hp         drat           wt
##    12.30337     -0.11144      0.01334     -0.02148      0.78711     -3.71530
##        qsec           vs           am         gear         carb
##     0.82104      0.31776      2.52023      0.65541     -0.19942
```

```r
lm(mpg ~ . - cyl, data = mtcars) # Theo tất cả các biến trừ biến cyl
```

```
##
## Call:
## lm(formula = mpg ~ . - cyl, data = mtcars)
##
## Coefficients:
## (Intercept)         disp           hp         drat           wt         qsec
##    10.96007      0.01283     -0.02191      0.83520     -3.69251      0.84244
##          vs           am         gear         carb
##     0.38975      2.57743      0.71155     -0.21958
```

```r
# Tìm mô hình HQTT phù hợp nhất dựa vào tiêu chuẩn ATC
# Hướng: forward, backward, both
# a. Forward: Xuất phát từ mô hình đầu tiên (mô hình đơn giản nhất),
# Ví dụ: Xét Y = a0 + e (Mô hình 0)
# -> Thu được một giá trị AIC
# -> Thêm 1 trong 10 biến vào (Mô hình 0): Y = a0 + a*X_i + e (i = 1, 2,..., 10)
# -> Thu được 10 mô hình đơn biến dạng Y = a0 + a1*X_i + e
# -> Chọn mô hình có tiêu chuẩn AIC nhỏ nhất
# -> Thêm 1 trong 9 biến còn lại vào mô hình đó
# -> Tiếp tục cho đến khi thu được mô hình Y = a0 + a1*X_1 + ... + am*X_m + e
# Khi thêm bất kì một biến nào vào mô hình trên mà giá trị AIC thu được
# đều lớn hơn giá trị AIC của mô hình đó thì dừng lại.
# Chọn mô hình đó là mô hình cuối cùng.

# Backward, Both: Tương tự
# Cách gọi hàm: step(object, scope, scale, direction)
# direction: ("forward", "backward", "both")
all <- lm(mpg ~ ., data = mtcars)
two <- lm(mpg ~ cyl + disp, data = mtcars)
fw <- step(two, formula(all), direction = "forward")
```

```
## Start:  AIC=74.33
## mpg ~ cyl + disp
##
##        Df Sum of Sq    RSS    AIC
## + wt    1    82.248 188.49 64.746
## + carb  1    30.232 240.51 72.545
## + am    1    18.659 252.08 74.048
## <none>              270.74 74.334
## + hp    1     9.371 261.37 75.206
## + drat  1     6.225 264.51 75.589
## + qsec  1     3.484 267.26 75.919
## + vs    1     1.076 269.66 76.206
## + gear  1     0.292 270.45 76.299
##
## Step:  AIC=64.75
## mpg ~ cyl + disp + wt
##
##        Df Sum of Sq    RSS    AIC
## + hp    1   18.0480 170.44 63.526
## + qsec  1   12.8240 175.67 64.492
## <none>              188.49 64.746
## + carb  1   11.3016 177.19 64.768
## + gear  1    2.6509 185.84 66.293
## + vs    1    0.9848 187.51 66.579
## + am    1    0.0666 188.43 66.735
## + drat  1    0.0004 188.49 66.746
##
## Step:  AIC=63.53
## mpg ~ cyl + disp + wt + hp
##
##        Df Sum of Sq    RSS    AIC
## <none>              170.44 63.526
## + am    1    7.3245 163.12 64.120
## + drat  1    3.0182 167.43 64.954
## + gear  1    2.7764 167.67 65.000
```

```
## + qsec  1    1.7590 168.69 65.194
## + vs    1    0.1554 170.29 65.496
## + carb  1    0.0009 170.44 65.525
```

```r
fw$anova
```

```
##   Step Df Deviance Resid. Df Resid. Dev      AIC
## 1      NA       NA        29   270.7403 74.33357
## 2 + wt -1 82.24790        28   188.4924 64.74629
## 3 + hp -1 18.04802        27   170.4444 63.52554
```

```r
summary(fw)
```

```
##
## Call:
## lm(formula = mpg ~ cyl + disp + wt + hp, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0562 -1.4636 -0.4281  1.2854  5.8269
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 40.82854    2.75747  14.807 1.76e-14 ***
## cyl         -1.29332    0.65588  -1.972 0.058947 .
## disp         0.01160    0.01173   0.989 0.331386
## wt          -3.85390    1.01547  -3.795 0.000759 ***
## hp          -0.02054    0.01215  -1.691 0.102379
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.513 on 27 degrees of freedom
## Multiple R-squared:  0.8486, Adjusted R-squared:  0.8262
## F-statistic: 37.84 on 4 and 27 DF,  p-value: 1.061e-10
```

```r
only <- lm(mpg ~ cyl, data = mtcars)

five <- lm(mpg ~ cyl + disp + hp + drat + wt, data = mtcars)
bw <- step(five, direction = "backward")
```

```
## Start:  AIC=64.95
## mpg ~ cyl + disp + hp + drat + wt
##
##        Df Sum of Sq    RSS    AIC
## - drat  1     3.018 170.44 63.526
## - disp  1     6.949 174.38 64.255
## <none>              167.43 64.954
## - cyl   1    15.411 182.84 65.772
## - hp    1    21.066 188.49 66.746
## - wt    1    77.476 244.90 75.124
##
## Step:  AIC=63.53
## mpg ~ cyl + disp + hp + wt
##
##        Df Sum of Sq    RSS    AIC
## - disp  1     6.176 176.62 62.665
## <none>              170.44 63.526
## - hp    1    18.048 188.49 64.746
## - cyl   1    24.546 194.99 65.831
```

```
## - wt    1    90.925 261.37 75.206
##
## Step:  AIC=62.66
## mpg ~ cyl + hp + wt
##
##        Df Sum of Sq    RSS    AIC
## <none>              176.62 62.665
## - hp    1    14.551 191.17 63.198
## - cyl   1    18.427 195.05 63.840
## - wt    1   115.354 291.98 76.750
```

```r
bw$anova
```

```
##     Step Df Deviance Resid. Df Resid. Dev      AIC
## 1        NA       NA        26   167.4261 64.95380
## 2 - drat  1 3.018242        27   170.4444 63.52554
## 3 - disp  1 6.176155        28   176.6205 62.66456
```

```r
summary(bw)
```

```
##
## Call:
## lm(formula = mpg ~ cyl + hp + wt, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9290 -1.5598 -0.5311  1.1850  5.8986
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 38.75179    1.78686  21.687  < 2e-16 ***
## cyl         -0.94162    0.55092  -1.709 0.098480 .
## hp          -0.01804    0.01188  -1.519 0.140015
## wt          -3.16697    0.74058  -4.276 0.000199 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.512 on 28 degrees of freedom
## Multiple R-squared:  0.8431, Adjusted R-squared:  0.8263
## F-statistic: 50.17 on 3 and 28 DF,  p-value: 2.184e-11
```

```r
bw <- step(all, direction = "backward")
```

```
## Start:  AIC=70.9
## mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb
##
##        Df Sum of Sq    RSS    AIC
## - cyl   1    0.0799 147.57 68.915
## - vs    1    0.1601 147.66 68.932
## - carb  1    0.4067 147.90 68.986
## - gear  1    1.3531 148.85 69.190
## - drat  1    1.6270 149.12 69.249
## - disp  1    3.9167 151.41 69.736
## - hp    1    6.8399 154.33 70.348
## - qsec  1    8.8641 156.36 70.765
## <none>              147.49 70.898
## - am    1   10.5467 158.04 71.108
## - wt    1   27.0144 174.51 74.280
##
```

```
## Step:  AIC=68.92
## mpg ~ disp + hp + drat + wt + qsec + vs + am + gear + carb
##
##        Df Sum of Sq    RSS    AIC
## - vs    1    0.2685 147.84 66.973
## - carb  1    0.5201 148.09 67.028
## - gear  1    1.8211 149.40 67.308
## - drat  1    1.9826 149.56 67.342
## - disp  1    3.9009 151.47 67.750
## - hp    1    7.3632 154.94 68.473
## <none>              147.57 68.915
## - qsec  1   10.0933 157.67 69.032
## - am    1   11.8359 159.41 69.384
## - wt    1   27.0280 174.60 72.297
##
## Step:  AIC=66.97
## mpg ~ disp + hp + drat + wt + qsec + am + gear + carb
##
##        Df Sum of Sq    RSS    AIC
## - carb  1    0.6855 148.53 65.121
## - gear  1    2.1437 149.99 65.434
## - drat  1    2.2139 150.06 65.449
## - disp  1    3.6467 151.49 65.753
## - hp    1    7.1060 154.95 66.475
## <none>              147.84 66.973
## - am    1   11.5694 159.41 67.384
## - qsec  1   15.6830 163.53 68.200
## - wt    1   27.3799 175.22 70.410
##
## Step:  AIC=65.12
## mpg ~ disp + hp + drat + wt + qsec + am + gear
##
##        Df Sum of Sq    RSS    AIC
## - gear  1     1.565 150.09 63.457
## - drat  1     1.932 150.46 63.535
## <none>              148.53 65.121
## - disp  1    10.110 158.64 65.229
## - am    1    12.323 160.85 65.672
## - hp    1    14.826 163.35 66.166
## - qsec  1    26.408 174.94 68.358
## - wt    1    69.127 217.66 75.350
##
## Step:  AIC=63.46
## mpg ~ disp + hp + drat + wt + qsec + am
##
##        Df Sum of Sq    RSS    AIC
## - drat  1     3.345 153.44 62.162
## - disp  1     8.545 158.64 63.229
## <none>              150.09 63.457
## - hp    1    13.285 163.38 64.171
## - am    1    20.036 170.13 65.466
## - qsec  1    25.574 175.67 66.491
## - wt    1    67.572 217.66 73.351
##
## Step:  AIC=62.16
## mpg ~ disp + hp + wt + qsec + am
```

```
##
##        Df Sum of Sq    RSS    AIC
## - disp  1     6.629 160.07 61.515
## <none>              153.44 62.162
## - hp    1    12.572 166.01 62.682
## - qsec  1    26.470 179.91 65.255
## - am    1    32.198 185.63 66.258
## - wt    1    69.043 222.48 72.051
##
## Step:  AIC=61.52
## mpg ~ hp + wt + qsec + am
##
##        Df Sum of Sq    RSS    AIC
## - hp    1     9.219 169.29 61.307
## <none>              160.07 61.515
## - qsec  1    20.225 180.29 63.323
## - am    1    25.993 186.06 64.331
## - wt    1    78.494 238.56 72.284
##
## Step:  AIC=61.31
## mpg ~ wt + qsec + am
##
##        Df Sum of Sq    RSS    AIC
## <none>              169.29 61.307
## - am    1    26.178 195.46 63.908
## - qsec  1   109.034 278.32 75.217
## - wt    1   183.347 352.63 82.790
```

```
bw$anova
```

```
##     Step Df   Deviance Resid. Df Resid. Dev      AIC
## 1         NA         NA        21   147.4944 70.89774
## 2  - cyl  1 0.07987121        22   147.5743 68.91507
## 3   - vs  1 0.26852280        23   147.8428 66.97324
## 4 - carb  1 0.68546077        24   148.5283 65.12126
## 5 - gear  1 1.56497053        25   150.0933 63.45667
## 6 - drat  1 3.34455117        26   153.4378 62.16190
## 7 - disp  1 6.62865369        27   160.0665 61.51530
## 8   - hp  1 9.21946935        28   169.2859 61.30730
```

```
summary(fw)
```

```
##
## Call:
## lm(formula = mpg ~ cyl + disp + wt + hp, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0562 -1.4636 -0.4281  1.2854  5.8269
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 40.82854    2.75747  14.807 1.76e-14 ***
## cyl         -1.29332    0.65588  -1.972 0.058947 .
## disp         0.01160    0.01173   0.989 0.331386
## wt          -3.85390    1.01547  -3.795 0.000759 ***
## hp          -0.02054    0.01215  -1.691 0.102379
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.513 on 27 degrees of freedom
## Multiple R-squared:  0.8486, Adjusted R-squared:  0.8262
## F-statistic: 37.84 on 4 and 27 DF,  p-value: 1.061e-10
```

```
both <- step(only, formula(all), direction = "both")
```

```
## Start:  AIC=76.49
## mpg ~ cyl
##
##        Df Sum of Sq     RSS     AIC
## + wt    1    117.16  191.17  63.198
## + disp  1     37.59  270.74  74.334
## + am    1     36.97  271.36  74.407
## <none>               308.33  76.494
## + hp    1     16.36  291.97  76.750
## + carb  1     16.17  292.16  76.770
## + drat  1     15.84  292.49  76.807
## + qsec  1     12.55  295.78  77.164
## + gear  1      5.43  302.90  77.926
## + vs    1      2.38  305.96  78.246
## - cyl   1    817.71 1126.05 115.943
##
## Step:  AIC=63.2
## mpg ~ cyl + wt
##
##        Df Sum of Sq    RSS    AIC
## + hp    1    14.551 176.62 62.665
## + carb  1    13.772 177.40 62.805
## <none>              191.17 63.198
## + qsec  1    10.567 180.60 63.378
## + gear  1     3.028 188.14 64.687
## + disp  1     2.680 188.49 64.746
## + vs    1     0.706 190.47 65.080
## + am    1     0.125 191.05 65.177
## + drat  1     0.001 191.17 65.198
## - cyl   1    87.150 278.32 73.217
## - wt    1   117.162 308.33 76.494
##
## Step:  AIC=62.66
## mpg ~ cyl + wt + hp
##
##        Df Sum of Sq    RSS    AIC
## <none>              176.62 62.665
## - hp    1    14.551 191.17 63.198
## + am    1     6.623 170.00 63.442
## + disp  1     6.176 170.44 63.526
## - cyl   1    18.427 195.05 63.840
## + carb  1     2.519 174.10 64.205
## + drat  1     2.245 174.38 64.255
## + qsec  1     1.401 175.22 64.410
## + gear  1     0.856 175.76 64.509
## + vs    1     0.060 176.56 64.654
## - wt    1   115.354 291.98 76.750
```

```r
both$anova
```

```
##   Step Df  Deviance Resid. Df Resid. Dev      AIC
## 1      NA        NA        30   308.3342 76.49435
## 2 + wt -1 117.16227        29   191.1720 63.19800
## 3 + hp -1  14.55145        28   176.6205 62.66456
```

```r
summary(both)
```

```
##
## Call:
## lm(formula = mpg ~ cyl + wt + hp, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9290 -1.5598 -0.5311  1.1850  5.8986
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 38.75179    1.78686  21.687  < 2e-16 ***
## cyl         -0.94162    0.55092  -1.709 0.098480 .
## wt          -3.16697    0.74058  -4.276 0.000199 ***
## hp          -0.01804    0.01188  -1.519 0.140015
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.512 on 28 degrees of freedom
## Multiple R-squared:  0.8431, Adjusted R-squared:  0.8263
## F-statistic: 50.17 on 3 and 28 DF,  p-value: 2.184e-11
```

```r
both <- step(five, direction = "both")
```

```
## Start:  AIC=64.95
## mpg ~ cyl + disp + hp + drat + wt
##
##        Df Sum of Sq    RSS    AIC
## - drat  1     3.018 170.44 63.526
## - disp  1     6.949 174.38 64.255
## <none>              167.43 64.954
## - cyl   1    15.411 182.84 65.772
## - hp    1    21.066 188.49 66.746
## - wt    1    77.476 244.90 75.124
##
## Step:  AIC=63.53
## mpg ~ cyl + disp + hp + wt
##
##        Df Sum of Sq    RSS    AIC
## - disp  1     6.176 176.62 62.665
## <none>              170.44 63.526
## - hp    1    18.048 188.49 64.746
## + drat  1     3.018 167.43 64.954
## - cyl   1    24.546 194.99 65.831
## - wt    1    90.925 261.37 75.206
##
## Step:  AIC=62.66
## mpg ~ cyl + hp + wt
##
##        Df Sum of Sq    RSS    AIC
```

```
## <none>              176.62 62.665
## - hp    1    14.551 191.17 63.198
## + disp  1     6.176 170.44 63.526
## - cyl   1    18.427 195.05 63.840
## + drat  1     2.245 174.38 64.255
## - wt    1   115.354 291.98 76.750
```

```r
both$anova
```

```
##     Step Df Deviance Resid. Df Resid. Dev      AIC
## 1        NA       NA        26   167.4261 64.95380
## 2 - drat  1 3.018242        27   170.4444 63.52554
## 3 - disp  1 6.176155        28   176.6205 62.66456
```

```r
summary(both)
```

```
##
## Call:
## lm(formula = mpg ~ cyl + hp + wt, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9290 -1.5598 -0.5311  1.1850  5.8986
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 38.75179    1.78686  21.687  < 2e-16 ***
## cyl         -0.94162    0.55092  -1.709 0.098480 .
## hp          -0.01804    0.01188  -1.519 0.140015
## wt          -3.16697    0.74058  -4.276 0.000199 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.512 on 28 degrees of freedom
## Multiple R-squared:  0.8431, Adjusted R-squared:  0.8263
## F-statistic: 50.17 on 3 and 28 DF,  p-value: 2.184e-11
```

```r
both <- step(all, direction = "both")
```

```
## Start:  AIC=70.9
## mpg ~ cyl + disp + hp + drat + wt + qsec + vs + am + gear + carb
##
##        Df Sum of Sq    RSS    AIC
## - cyl   1    0.0799 147.57 68.915
## - vs    1    0.1601 147.66 68.932
## - carb  1    0.4067 147.90 68.986
## - gear  1    1.3531 148.85 69.190
## - drat  1    1.6270 149.12 69.249
## - disp  1    3.9167 151.41 69.736
## - hp    1    6.8399 154.33 70.348
## - qsec  1    8.8641 156.36 70.765
## <none>              147.49 70.898
## - am    1   10.5467 158.04 71.108
## - wt    1   27.0144 174.51 74.280
##
## Step:  AIC=68.92
## mpg ~ disp + hp + drat + wt + qsec + vs + am + gear + carb
##
##        Df Sum of Sq    RSS    AIC
```

```
## - vs    1    0.2685 147.84 66.973
## - carb  1    0.5201 148.09 67.028
## - gear  1    1.8211 149.40 67.308
## - drat  1    1.9826 149.56 67.342
## - disp  1    3.9009 151.47 67.750
## - hp    1    7.3632 154.94 68.473
## <none>              147.57 68.915
## - qsec  1   10.0933 157.67 69.032
## - am    1   11.8359 159.41 69.384
## + cyl   1    0.0799 147.49 70.898
## - wt    1   27.0280 174.60 72.297
##
## Step:  AIC=66.97
## mpg ~ disp + hp + drat + wt + qsec + am + gear + carb
##
##        Df Sum of Sq    RSS    AIC
## - carb  1    0.6855 148.53 65.121
## - gear  1    2.1437 149.99 65.434
## - drat  1    2.2139 150.06 65.449
## - disp  1    3.6467 151.49 65.753
## - hp    1    7.1060 154.95 66.475
## <none>              147.84 66.973
## - am    1   11.5694 159.41 67.384
## - qsec  1   15.6830 163.53 68.200
## + vs    1    0.2685 147.57 68.915
## + cyl   1    0.1883 147.66 68.932
## - wt    1   27.3799 175.22 70.410
##
## Step:  AIC=65.12
## mpg ~ disp + hp + drat + wt + qsec + am + gear
##
##        Df Sum of Sq    RSS    AIC
## - gear  1     1.565 150.09 63.457
## - drat  1     1.932 150.46 63.535
## <none>              148.53 65.121
## - disp  1    10.110 158.64 65.229
## - am    1    12.323 160.85 65.672
## - hp    1    14.826 163.35 66.166
## + carb  1     0.685 147.84 66.973
## + vs    1     0.434 148.09 67.028
## + cyl   1     0.414 148.11 67.032
## - qsec  1    26.408 174.94 68.358
## - wt    1    69.127 217.66 75.350
##
## Step:  AIC=63.46
## mpg ~ disp + hp + drat + wt + qsec + am
##
##        Df Sum of Sq    RSS    AIC
## - drat  1     3.345 153.44 62.162
## - disp  1     8.545 158.64 63.229
## <none>              150.09 63.457
## - hp    1    13.285 163.38 64.171
## + gear  1     1.565 148.53 65.121
## + cyl   1     1.003 149.09 65.242
## + vs    1     0.645 149.45 65.319
## + carb  1     0.107 149.99 65.434
```

```
## - am    1    20.036 170.13 65.466
## - qsec  1    25.574 175.67 66.491
## - wt    1    67.572 217.66 73.351
##
## Step:  AIC=62.16
## mpg ~ disp + hp + wt + qsec + am
##
##        Df Sum of Sq    RSS    AIC
## - disp  1     6.629 160.07 61.515
## <none>              153.44 62.162
## - hp    1    12.572 166.01 62.682
## + drat  1     3.345 150.09 63.457
## + gear  1     2.977 150.46 63.535
## + cyl   1     2.447 150.99 63.648
## + vs    1     1.121 152.32 63.927
## + carb  1     0.011 153.43 64.160
## - qsec  1    26.470 179.91 65.255
## - am    1    32.198 185.63 66.258
## - wt    1    69.043 222.48 72.051
##
## Step:  AIC=61.52
## mpg ~ hp + wt + qsec + am
##
##        Df Sum of Sq    RSS    AIC
## - hp    1     9.219 169.29 61.307
## <none>              160.07 61.515
## + disp  1     6.629 153.44 62.162
## + carb  1     3.227 156.84 62.864
## + drat  1     1.428 158.64 63.229
## - qsec  1    20.225 180.29 63.323
## + cyl   1     0.249 159.82 63.465
## + vs    1     0.249 159.82 63.466
## + gear  1     0.171 159.90 63.481
## - am    1    25.993 186.06 64.331
## - wt    1    78.494 238.56 72.284
##
## Step:  AIC=61.31
## mpg ~ wt + qsec + am
##
##        Df Sum of Sq    RSS    AIC
## <none>              169.29 61.307
## + hp    1     9.219 160.07 61.515
## + carb  1     8.036 161.25 61.751
## + disp  1     3.276 166.01 62.682
## + cyl   1     1.501 167.78 63.022
## + drat  1     1.400 167.89 63.042
## + gear  1     0.123 169.16 63.284
## + vs    1     0.000 169.29 63.307
## - am    1    26.178 195.46 63.908
## - qsec  1   109.034 278.32 75.217
## - wt    1   183.347 352.63 82.790
```

```
both$anova
```

```
##     Step Df   Deviance Resid. Df Resid. Dev      AIC
## 1        NA         NA        21   147.4944 70.89774
## 2  - cyl  1 0.07987121        22   147.5743 68.91507
```

```
## 3   - vs  1 0.26852280        23   147.8428 66.97324
## 4 - carb  1 0.68546077        24   148.5283 65.12126
## 5 - gear  1 1.56497053        25   150.0933 63.45667
## 6 - drat  1 3.34455117        26   153.4378 62.16190
## 7 - disp  1 6.62865369        27   160.0665 61.51530
## 8   - hp  1 9.21946935        28   169.2859 61.30730
```

```r
summary(both)
```

```
##
## Call:
## lm(formula = mpg ~ wt + qsec + am, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4811 -1.5555 -0.7257  1.4110  4.6610
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.6178     6.9596   1.382 0.177915
## wt           -3.9165     0.7112  -5.507 6.95e-06 ***
## qsec          1.2259     0.2887   4.247 0.000216 ***
## am            2.9358     1.4109   2.081 0.046716 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.459 on 28 degrees of freedom
## Multiple R-squared:  0.8497, Adjusted R-squared:  0.8336
## F-statistic: 52.75 on 3 and 28 DF,  p-value: 1.21e-11
```